

SKILL

WEEK-4

2520090046

1. CREATE CUSTOMER TABLE

```
1 • CREATE DATABASE IF NOT EXISTS sbiDB;
2 • USE sbiDB;
3 • CREATE TABLE Customer (
4     Customer_ID INT PRIMARY KEY,
5     Customer_Name VARCHAR(100) NOT NULL,
6     Phone VARCHAR(15),
7     Email VARCHAR(100),
8     City VARCHAR(50)
9 );
```

Output				
Action Output				
#	Time	Action	Message	Duration / Fetch
✓ 1	12:23:50	USE BankDB	0 row(s) affected	0.000 sec
✓ 2	12:24:22	CREATE DATABASE IF NOT EXISTS sbiDB	1 row(s) affected	0.000 sec
✓ 3	12:24:22	USE sbiDB	0 row(s) affected	0.000 sec
✓ 4	12:25:24	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT NULL...	0 row(s) affected	0.016 sec

2. CREATE ACCOUNT TABLE

```
10 • CREATE TABLE Account (
11     Account_No INT PRIMARY KEY,
12     Customer_ID INT,
13     Account_Type VARCHAR(20),
14     Balance DECIMAL(12,2) DEFAULT 0,
15     Branch VARCHAR(50),
16
17     FOREIGN KEY (Customer_ID)
18     REFERENCES Customer(Customer_ID)
19 );
20
21
```

Output				
Action Output				
#	Time	Action	Message	Duration / Fetch
✓ 1	12:23:50	USE BankDB	0 row(s) affected	0.000 sec
✓ 2	12:24:22	CREATE DATABASE IF NOT EXISTS sbiDB	1 row(s) affected	0.000 sec
✓ 3	12:24:22	USE sbiDB	0 row(s) affected	0.000 sec
✓ 4	12:25:24	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT NULL...	0 row(s) affected	0.016 sec
✓ 5	12:25:57	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Account_Type VARCHA...	0 row(s) affected	0.032 sec

3. CREATE TRANSACTION TABLE

```
20 • CREATE TABLE Account (
21     Account_No INT PRIMARY KEY,
22     Customer_ID INT,
23     Account_Type VARCHAR(20),
24     Balance DECIMAL(12,2) DEFAULT 0,
25     Branch VARCHAR(50),
26
27     FOREIGN KEY (Customer_ID)
28     REFERENCES Customer(Customer_ID)
29 );
30
31
```

Output				
Action Output				
#	Time	Action	Message	Duration / Fetch
✓ 1	12:23:50	USE BankDB	0 row(s) affected	0.000 sec
✓ 2	12:24:22	CREATE DATABASE IF NOT EXISTS sbiDB	1 row(s) affected	0.000 sec
✓ 3	12:24:22	USE sbiDB	0 row(s) affected	0.000 sec
✓ 4	12:25:24	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT NULL...	0 row(s) affected	0.016 sec
✓ 5	12:25:57	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Account_Type VARCHA...	0 row(s) affected	0.032 sec
✗ 6	12:27:07	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Account_Type VARCHA...	Error Code: 1050. Table 'account' already exists	0.000 sec

4. CREATE TRANSACTION TABLE

```

30  CREATE TABLE Bank_Transaction (
31      Transaction_ID INT PRIMARY KEY AUTO_INCREMENT,
32      Account_No INT,
33      Transaction_Type VARCHAR(20),
34      Amount DECIMAL(12,2),
35      Transaction_Date DATETIME DEFAULT CURRENT_TIMESTAMP,
36
37      FOREIGN KEY (Account_No)
38      REFERENCES Account(Account_No)
39  );
40

```

Output

#	Time	Action	Message	Duration / Fetch
2	12:24:22	CREATE DATABASE IF NOT EXISTS sbiDB	1 row(s) affected	0.000 sec
3	12:24:22	USE sbiDB	0 row(s) affected	0.000 sec
4	12:25:24	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT NU...	0 row(s) affected	0.016 sec
5	12:25:57	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Account_Type VARCH...	0 row(s) affected	0.032 sec
6	12:27:07	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Account_Type VARCH...	Error Code: 1050. Table 'account' already exists	0.000 sec
7	12:28:15	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_INCREMENT, Account_...	0 row(s) affected	0.031 sec

5. CREATE LOAN TABLE

```

40  CREATE TABLE Loan (
41      Loan_ID INT PRIMARY KEY,
42      Customer_ID INT,
43      Loan_Type VARCHAR(30),
44      Loan_Amount DECIMAL(12,2),
45      Interest_Rate DECIMAL(5,2),
46
47      FOREIGN KEY (Customer_ID)
48      REFERENCES Customer(Customer_ID)
49  );
50

```

Output

#	Time	Action	Message	Duration / Fetch
3	12:24:22	USE sbiDB	0 row(s) affected	0.000 sec
4	12:25:24	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT NU...	0 row(s) affected	0.016 sec
5	12:25:57	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Account_Type VARCH...	0 row(s) affected	0.032 sec
6	12:27:07	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Account_Type VARCH...	Error Code: 1050. Table 'account' already exists	0.000 sec
7	12:28:15	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_INCREMENT, Account_...	0 row(s) affected	0.031 sec
8	12:28:56	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, Customer_ID INT, Loan_Type VARCHAR(30), ...	0 row(s) affected	0.031 sec

6. INSERT CUSTOMER DATA

```

50  INSERT INTO Customer
51  (Customer_ID, Customer_Name, Phone, Email, City)
52  VALUES
53  (101, 'Ravi Kumar', '9876543210', 'ravi@gmail.com', 'Hyderabad'),
54
55  (102, 'Priya Sharma', '9876543211', 'priya@gmail.com', 'Vijayawada'),
56
57  (103, 'Arjun Reddy', '9876543212', 'arjun@gmail.com', 'Bangalore'),
58
59  (104, 'Sneha Rao', '9876543213', 'sneha@gmail.com', 'Chennai'),
60
61  (105, 'Kiran Kumar', '9876543214', 'kiran@gmail.com', 'Hyderabad');
62

```

Output

#	Time	Action	Message	Duration / Fetch
4	12:25:24	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT NU...	0 row(s) affected	0.016 sec
5	12:25:57	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Account_Type VARCH...	0 row(s) affected	0.032 sec
6	12:27:07	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Account_Type VARCH...	Error Code: 1050. Table 'account' already exists	0.000 sec
7	12:28:15	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_INCREMENT, Account_...	0 row(s) affected	0.031 sec
8	12:28:56	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, Customer_ID INT, Loan_Type VARCHAR(30), ...	0 row(s) affected	0.031 sec
9	12:29:29	INSERT INTO Customer (Customer_ID, Customer_Name, Phone, Email, City) VALUES (101, 'Ravi Kumar', '9876...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.000 sec

7. INSERT ACCOUNT DATA

```

62 • INSERT INTO Account
63 (Account_No, Customer_ID, Account_Type, Balance, Branch)
64 VALUES
65 (10001, 101, 'Savings', 50000, 'Hyderabad'),
66 (10002, 102, 'Savings', 75000, 'Vijayawada'),
67 (10003, 103, 'Current', 120000, 'Bangalore'),
68 (10004, 104, 'Savings', 45000, 'Chennai'),
69 (10005, 105, 'Current', 90000, 'Hyderabad');
70
71
72
73
74
75

```

Output				
Action Output				
#	Time	Action	Message	Duration / Fetch
5	12:25:57	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Account_Type VARCHAR(30), Balance INT, Branch VARCHAR(30))	0 row(s) affected	0.032 sec
6	12:27:07	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Account_Type VARCHAR(30), Balance INT, Branch VARCHAR(30))	Error Code: 1050. Table 'account' already exists	0.000 sec
7	12:28:15	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_INCREMENT, Account_No INT, Transaction_Type VARCHAR(30), Amount INT)	0 row(s) affected	0.031 sec
8	12:28:56	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, Customer_ID INT, Loan_Type VARCHAR(30), Loan_Amount INT, Interest_Rate DECIMAL(5,2))	0 row(s) affected	0.031 sec
9	12:29:29	INSERT INTO Customer (Customer_ID, Customer_Name, Phone, Email, City) VALUES (101, 'Ravi Kumar', '9876543210', 'ravi.kumar@gmail.com', 'Hyderabad')	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.000 sec
10	12:30:07	INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branch) VALUES (10001, 101, 'Savings', 50000, 'Hyderabad')	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.000 sec

8. INSERT TRANSACTION DATA

```

75 • INSERT INTO Bank_Transaction
76 (Account_No, Transaction_Type, Amount)
77 VALUES
78 (10001, 'DEPOSIT', 10000),
79 (10002, 'DEPOSIT', 15000),
80 (10003, 'WITHDRAW', 20000),
81 (10004, 'DEPOSIT', 5000),
82 (10005, 'WITHDRAW', 10000);
83
84
85
86
87

```

Output				
Action Output				
#	Time	Action	Message	Duration / Fetch
6	12:27:07	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Account_Type VARCHAR(30), Balance INT, Branch VARCHAR(30))	Error Code: 1050. Table 'account' already exists	0.000 sec
7	12:28:15	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_INCREMENT, Account_No INT, Transaction_Type VARCHAR(30), Amount INT)	0 row(s) affected	0.031 sec
8	12:28:56	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, Customer_ID INT, Loan_Type VARCHAR(30), Loan_Amount INT, Interest_Rate DECIMAL(5,2))	0 row(s) affected	0.031 sec
9	12:29:29	INSERT INTO Customer (Customer_ID, Customer_Name, Phone, Email, City) VALUES (101, 'Ravi Kumar', '9876543210', 'ravi.kumar@gmail.com', 'Hyderabad')	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.000 sec
10	12:30:07	INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branch) VALUES (10001, 101, 'Savings', 50000, 'Hyderabad')	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.000 sec
11	12:30:55	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', 10000)	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.000 sec

9. INSERT LOAN DATA

```

87 • INSERT INTO Loan
88 (Loan_ID, Customer_ID, Loan_Type, Loan_Amount, Interest_Rate)
89 VALUES
90 (501, 101, 'Home Loan', 5000000, 7.5),
91 (502, 102, 'Education Loan', 1000000, 6.5),
92 (503, 103, 'Car Loan', 800000, 8.2),
93 (504, 104, 'Personal Loan', 500000, 10.5);
94
95
96
97

```

Output				
Action Output				
#	Time	Action	Message	Duration / Fetch
7	12:28:15	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_INCREMENT, Account_No INT, Transaction_Type VARCHAR(30), Amount INT)	0 row(s) affected	0.031 sec
8	12:28:56	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, Customer_ID INT, Loan_Type VARCHAR(30), Loan_Amount INT, Interest_Rate DECIMAL(5,2))	0 row(s) affected	0.031 sec
9	12:29:29	INSERT INTO Customer (Customer_ID, Customer_Name, Phone, Email, City) VALUES (101, 'Ravi Kumar', '9876543210', 'ravi.kumar@gmail.com', 'Hyderabad')	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.000 sec
10	12:30:07	INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branch) VALUES (10001, 101, 'Savings', 50000, 'Hyderabad')	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.000 sec
11	12:30:55	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', 10000)	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.000 sec
12	12:31:24	INSERT INTO Loan (Loan_ID, Customer_ID, Loan_Type, Loan_Amount, Interest_Rate) VALUES (501, 101, 'Home Loan', 5000000, 7.5)	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0	0.015 sec

10. DISPLAY TABLE DATA

```

97 • SELECT * FROM Customer;
98
99 • SELECT * FROM Account;
100
101 • SELECT * FROM Bank_Transaction;
102
103 • SELECT * FROM Loan;

```

Loan_ID	Customer_ID	Loan_Type	Loan_Amount	Interest_Rate
501	101	Home Loan	5000000.00	7.50
502	102	Education Loan	1000000.00	6.50
503	103	Car Loan	800000.00	8.20
504	104	Personal Loan	500000.00	10.50

#	Time	Action	Message	Duration / Fetch
8	12:28:56	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, Customer_ID INT, Loan_Type VARCHAR(30), ...	0 row(s) affected	0.031 sec
9	12:29:29	INSERT INTO Customer (Customer_ID, Customer_Name, Phone, Email, City) VALUES (101, 'Ravi Kumar', '987...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.000 sec
10	12:30:07	INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branch) VALUES (10001, 101, '...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.000 sec
11	12:30:55	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', 100...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.000 sec
12	12:31:24	INSERT INTO Loan (Loan_ID, Customer_ID, Loan_Type, Loan_Amount, Interest_Rate) VALUES (501, 101, 'H...	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0	0.015 sec
13	12:31:59	SELECT * FROM Loan LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec

11. PROCEDURE TO DISPLAY ALL CUSTOMERS

```

105 • CREATE PROCEDURE GetAllCustomers()
106     SELECT * FROM Customer;
107 DELIMITER ;
108 • CALL GetAllCustomers();
109

```

Customer_ID	Customer_Name	Phone	Email	City
101	Ravi Kumar	9876543210	ravi@gmail.com	Hyderabad
102	Priya Sharma	9876543211	priya@gmail.com	Vijayawada
103	Arjun Reddy	9876543212	arjun@gmail.com	Bangalore
104	Sneha Rao	9876543213	sneha@gmail.com	Chennai
105	Kiran Kumar	9876543214	kiran@gmail.com	Hyderabad

#	Time	Action	Message	Duration / Fetch
13	12:31:59	SELECT * FROM Loan LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec
14	12:33:01	CREATE PROCEDURE GetAllCustomers()	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL ser...	0.000 sec
15	12:33:06	SELECT * FROM Customer LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
16	12:33:19	CALL GetAllCustomers()	Error Code: 1305. PROCEDURE sbidb.GetAllCustomers does not exist	0.000 sec
17	12:33:54	CREATE PROCEDURE GetAllCustomers() SELECT * FROM Customer	0 row(s) affected	0.000 sec
18	12:34:00	CALL GetAllCustomers()	5 row(s) returned	0.000 sec / 0.000 sec

12. PROCEDURE TO DISPLAY ACCOUNT DETAILS FOR A GIVEN ACCOUNT NUMBER

```

109 • CREATE PROCEDURE GetAccountDetails(
110     IN p_Account_No INT
111 )
112     SELECT *
113     FROM Account
114     WHERE Account_No = p_Account_No;
115
116 • CALL GetAccountDetails(10001);
117

```

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	50000.00	Hyderabad

#	Time	Action	Message	Duration / Fetch
19	12:35:20	SELECT * FROM Account WHERE Account_No = p_Account_No LIMIT 0, 1000	Error Code: 1054. Unknown column 'p_Account_No' in 'where clause'	0.000 sec
20	12:35:28	CALL GetAccountDetails(10001)	Error Code: 1305. PROCEDURE sbidb.GetAccountDetails does not exist	0.000 sec
21	12:35:50	SELECT * FROM Account WHERE Account_No = p_Account_No LIMIT 0, 1000	Error Code: 1054. Unknown column 'p_Account_No' in 'where clause'	0.000 sec
22	12:35:51	SELECT * FROM Account WHERE Account_No = p_Account_No LIMIT 0, 1000	Error Code: 1054. Unknown column 'p_Account_No' in 'where clause'	0.000 sec
23	12:36:00	CREATE PROCEDURE GetAccountDetails(IN p_Account_No INT) SELECT * FROM Account WH...	0 row(s) affected	0.016 sec
24	12:36:04	CALL GetAccountDetails(10001)	1 row(s) returned	0.000 sec / 0.000 sec

13. PROCEDURE TO FIND CUSTOMER ACCOUNTS

```

122     C.Customer_Name,
123     A.Account_No,
124     A.Account_Type,
125     A.Balance,
126     A.Branch
127 FROM Customer C
128 JOIN Account A
129 ON C.Customer_ID = A.Customer_ID
130 WHERE C.Customer_ID = p_Customer_ID;
131
132 • CALL GetCustomerAccounts(101);
133

```

Customer_ID	Customer_Name	Account_No	Account_Type	Balance	Branch
101	Ravi Kumar	10001	Savings	50000.00	Hyderabad

Result 5 x Read Only

Output

#	Time	Action	Message	Duration / Fetch
25	12:37:08	CALL GetCustomerAccounts(101)	Error Code: 1305. PROCEDURE sbidb.GetCustomerAccounts does not exist	0.015 sec
26	12:37:16	CREATE PROCEDURE GetCustomerAccounts(IN p_Customer_ID INT) SELECT C.Customer_ID, ...	0 row(s) affected	0.000 sec
27	12:37:20	CREATE PROCEDURE GetCustomerAccounts(IN p_Customer_ID INT) SELECT C.Customer_ID, ...	Error Code: 1304. PROCEDURE GetCustomerAccounts already exists	0.000 sec
28	12:37:25	FROM Customer C JOIN Account A ON C.Customer_ID = A.Customer_ID WHERE C.Customer_ID = p_...	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL ser...	0.000 sec
29	12:37:33	CREATE PROCEDURE GetCustomerAccounts(IN p_Customer_ID INT) SELECT C.Customer_ID, ...	Error Code: 1304. PROCEDURE GetCustomerAccounts already exists	0.000 sec
30	12:37:41	CALL GetCustomerAccounts(101)	1 row(s) returned	0.000 sec / 0.000 sec

14. PROCEDURE TO DEPOSIT MONEY

```

133 • CREATE PROCEDURE DepositMoney(
134     IN p_Account_No INT,
135     IN p_Amount DECIMAL(12,2)
136 )
137 UPDATE Account
138 SET Balance = Balance + p_Amount
139 WHERE Account_No = p_Account_No;
140
141 • CALL DepositMoney(10001, 5000);
142
143
144

```

Output

#	Time	Action	Message	Duration / Fetch
30	12:37:41	CALL GetCustomerAccounts(101)	1 row(s) returned	0.000 sec / 0.000 sec
31	12:38:39	CREATE DATABASE IF NOT EXISTS sbidb	1 row(s) affected, 1 warning(s): 1007 Can't create database 'sbidb'; database exists	0.000 sec
32	12:38:39	USE sbidb	0 row(s) affected	0.000 sec
33	12:38:39	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT NU...	Error Code: 1050. Table 'customer' already exists	0.016 sec
34	12:38:49	CREATE PROCEDURE DepositMoney(IN p_Account_No INT, IN p_Amount DECIMAL(12,2)) UPDAT...	0 row(s) affected	0.000 sec
35	12:38:53	CALL DepositMoney(10001, 5000)	1 row(s) affected	0.000 sec

15. PROCEDURE TO WITHDRAW MONEY

```

142 • CREATE PROCEDURE WithdrawMoney(
143     IN p_Account_No INT,
144     IN p_Amount DECIMAL(12,2)
145 )
146
147 UPDATE Account
148 SET Balance = Balance - p_Amount
149 WHERE Account_No = p_Account_No;
150
151 • CALL WithdrawMoney(10001, 3000);
152

```

Output

#	Time	Action	Message	Duration / Fetch
32	12:38:39	USE sbidb	0 row(s) affected	0.000 sec
33	12:38:39	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT NU...	Error Code: 1050. Table 'customer' already exists	0.016 sec
34	12:38:49	CREATE PROCEDURE DepositMoney(IN p_Account_No INT, IN p_Amount DECIMAL(12,2)) UPDAT...	0 row(s) affected	0.000 sec
35	12:38:53	CALL DepositMoney(10001, 5000)	1 row(s) affected	0.000 sec
36	12:39:52	CREATE PROCEDURE WithdrawMoney(IN p_Account_No INT, IN p_Amount DECIMAL(12,2)) UPD...	0 row(s) affected	0.000 sec
37	12:39:56	CALL WithdrawMoney(10001, 3000)	1 row(s) affected	0.016 sec

16. TRIGGER – PREVENT NEGATIVE BALANCE

152	•	DROP TRIGGER IF EXISTS CheckBalance;			
153					
154		DELIMITER //			
155					
156	•	CREATE TRIGGER CheckBalance			
157		BEFORE UPDATE ON Account			
158		FOR EACH ROW			
159		BEGIN			
160		IF NEW.Balance < 0 THEN			
161		SIGNAL SQLSTATE '45000'			
162		SET MESSAGE_TEXT = 'Transaction failed: Insufficient balance';			
163		END IF;			
164		END //			
165					
166		DELIMITER ;			

#	Time	Action	Message	Duration / Fetch
51	13:37:20	CREATE TRIGGER CheckBalance BEFORE UPDATE ON Account FOR EACH ROW BEGIN	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL ser...	0.000 sec
52	13:44:12	CREATE TRIGGER CheckBalance BEFORE UPDATE ON Account FOR EACH ROW BEGIN	IF NEW.Bala... Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL ser...	0.016 sec
53	13:44:18	CREATE TRIGGER CheckBalance BEFORE UPDATE ON Account FOR EACH ROW BEGIN	IF NEW.Bala... Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL ser...	0.000 sec
54	13:49:52	CREATE TRIGGER CheckBalance BEFORE UPDATE ON Account FOR EACH ROW BEGIN	IF NEW.Bala... Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL ser...	0.000 sec
55	13:49:56	CREATE TRIGGER CheckBalance BEFORE UPDATE ON Account FOR EACH ROW BEGIN	IF NEW.Bala... Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL ser...	0.000 sec
56	13:50:33	CREATE TRIGGER CheckBalance BEFORE UPDATE ON Account FOR EACH ROW BEGIN	IF NEW.Bala... 0 row(s) affected	0.000 sec

17. TRIGGER – PREVENT NEGATIVE DEPOSIT

167	•	DROP TRIGGER IF EXISTS CheckTransactionAmount;			
168					
169		DELIMITER //			
170					
171	•	CREATE TRIGGER CheckTransactionAmount			
172		BEFORE INSERT ON Bank_Transaction			
173		FOR EACH ROW			
174		BEGIN			
175		IF NEW.Amount <= 0 THEN			
176		SIGNAL SQLSTATE '45000'			
177		SET MESSAGE_TEXT = 'Transaction amount must be greater than zero';			
178		END IF;			
179		END //			
180					
181		DELIMITER ;			
182	•	INSERT INTO Bank_Transaction			
183		(Account_No, Transaction_Type, Amount)			
184		VALUES			
185		(10001, 'DEPOSIT', 5000);			

#	Time	Action	Message	Duration / Fetch
55	13:49:56	CREATE TRIGGER CheckBalance BEFORE UPDATE ON Account FOR EACH ROW BEGIN	IF NEW.Bala... Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL ser...	0.000 sec
56	13:50:33	CREATE TRIGGER CheckBalance BEFORE UPDATE ON Account FOR EACH ROW BEGIN	IF NEW.Bala... 0 row(s) affected	0.000 sec
57	13:52:29	CREATE TRIGGER CheckTransactionAmount BEFORE INSERT ON Bank_Transaction FOR EACH ROW BE...	0 row(s) affected	0.000 sec
58	13:52:38	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', -50...	Error Code: 1644. Transaction amount must be greater than zero	0.000 sec
59	13:52:46	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', -50...	Error Code: 1644. Transaction amount must be greater than zero	0.000 sec
60	13:52:55	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', 5000)	1 row(s) affected	0.000 sec

18. CREATE TRANSACTION AUDIT TABLE

186	•	CREATE TABLE Transaction_Audit (			
187		Audit_ID INT PRIMARY KEY AUTO_INCREMENT,			
188		Transaction_ID INT,			
189		Account_No INT,			
190		Transaction_Type VARCHAR(20),			
191		Amount DECIMAL(12,2),			
192		Audit_Date DATETIME DEFAULT CURRENT_TIMESTAMP			
193		);			
194					
195					

#	Time	Action	Message	Duration / Fetch
56	13:50:33	CREATE TRIGGER CheckBalance BEFORE UPDATE ON Account FOR EACH ROW BEGIN	IF NEW.Bala... 0 row(s) affected	0.000 sec
57	13:52:29	CREATE TRIGGER CheckTransactionAmount BEFORE INSERT ON Bank_Transaction FOR EACH ROW BE...	0 row(s) affected	0.000 sec
58	13:52:38	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', -50...	Error Code: 1644. Transaction amount must be greater than zero	0.000 sec
59	13:52:46	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', -50...	Error Code: 1644. Transaction amount must be greater than zero	0.000 sec
60	13:52:55	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', 5000)	1 row(s) affected	0.000 sec
61	13:53:38	CREATE TABLE Transaction_Audit (Audit_ID INT PRIMARY KEY AUTO_INCREMENT, Transaction_ID ...	0 row(s) affected	0.016 sec

19. TRIGGER – TRANSACTION AUDIT

```

215     );
216 END //
217 DELIMITER ;
218 -- Step 1: Insert record (Triggers Audit)
219 INSERT INTO Bank_Transaction
220 (Account_No, Transaction_Type, Amount)
221 VALUES
222 (10001, 'DEPOSIT', 2500);
223
224 -- Step 2: Check audit log
225 SELECT * FROM Transaction_Audit;

```

Audit_ID	Transaction_ID	Account_No	Transaction_Type	Amount	Audit_Date
60	10001	10001	DEPOSIT	2500	2023-10-27 13:52:55

Transaction_Audit 6 x

#	Time	Action	Message	Duration / Fetch
58	13:52:38	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', -5000)	Error Code: 1644. Transaction amount must be greater than zero	0.000 sec
59	13:52:46	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', -5000)	Error Code: 1644. Transaction amount must be greater than zero	0.000 sec
60	13:52:55	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', 2500)	1 row(s) affected	0.000 sec
61	13:53:38	CREATE TABLE Transaction_Audit (Audit_ID INT PRIMARY KEY AUTO_INCREMENT, Transaction_ID INT)	0 row(s) affected	0.016 sec
62	13:54:36	CREATE TRIGGER TransactionAudit AFTER INSERT ON Bank_Transaction FOR EACH ROW BEGIN INSERT INTO Transaction_Audit (Audit_ID, Transaction_ID) VALUES (LAST_INSERT_ID(), @@transaction_id);	0 row(s) affected	0.000 sec
63	13:55:50	SELECT * FROM Transaction_Audit LIMIT 0, 1000	0 row(s) returned	0.000 sec / 0.000 sec

20. TRIGGER – AUTOMATICALLY UPDATE ACCOUNT BALANCE

```

246 DELIMITER ;
247 -- Step 1: Insert record
248 INSERT INTO Bank_Transaction
249 (Account_No, Transaction_Type, Amount)
250 VALUES
251 (10001, 'DEPOSIT', 5000);
252
253 -- Step 2: Check updated balance
254 SELECT *
255 FROM Account
256 WHERE Account_No = 10001;

```

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	52000.00	Hyderabad

Account 7 x

#	Time	Action	Message	Duration / Fetch
60	13:52:55	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', 5000)	1 row(s) affected	0.000 sec
61	13:53:38	CREATE TABLE Transaction_Audit (Audit_ID INT PRIMARY KEY AUTO_INCREMENT, Transaction_ID INT)	0 row(s) affected	0.016 sec
62	13:54:36	CREATE TRIGGER TransactionAudit AFTER INSERT ON Bank_Transaction FOR EACH ROW BEGIN INSERT INTO Transaction_Audit (Audit_ID, Transaction_ID) VALUES (LAST_INSERT_ID(), @@transaction_id);	0 row(s) affected	0.000 sec
63	13:55:50	SELECT * FROM Transaction_Audit LIMIT 0, 1000	0 row(s) returned	0.000 sec / 0.000 sec
64	13:56:31	CREATE TRIGGER UpdateBalanceAfterTransaction AFTER INSERT ON Bank_Transaction FOR EACH ROW BEGIN UPDATE Account SET Balance = Balance + @@amount WHERE Account_No = @@account_no;	0 row(s) affected	0.000 sec
65	13:56:40	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec

21. TRIGGER – PREVENT WITHDRAWAL ABOVE BALANCE

```

268     INTO CurrentBalance
269     FROM Account
270     WHERE Account_No = NEW.Account_No;
271
272     IF NEW.Transaction_Type = 'WITHDRAW'
273     AND NEW.Amount > CurrentBalance THEN
274
275         SIGNAL SQLSTATE '45000'
276         SET MESSAGE_TEXT = 'Withdrawal failed: Insufficient balance';
277     END IF;
278 END //
279
280 DELIMITER ;
281 • INSERT INTO Bank_Transaction
282 (Account_No, Transaction_Type, Amount)
283 VALUES
284 • (10001, 'WITHDRAW', 100);INSERT INTO Bank_Transaction
285 (Account_No, Transaction_Type, Amount)
286 VALUES
287 (10001, 'WITHDRAW', 100);

```

Output

#	Time	Action	Message	Duration / Fetch
64	13:56:31	CREATE TRIGGER UpdateBalanceAfterTransaction AFTER INSERT ON Bank_Transaction FOR EACH ROW...	0 row(s) affected	0.000 sec
65	13:56:40	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
66	13:57:34	CREATE TRIGGER PreventInsufficientWithdrawal BEFORE INSERT ON Bank_Transaction FOR EACH ROW...	0 row(s) affected	0.016 sec
67	13:57:42	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'WITHDRAW', ...	Error Code: 1644. Withdrawal failed: Insufficient balance	0.000 sec
68	13:57:49	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'WITHDRAW', ...	Error Code: 1644. Withdrawal failed: Insufficient balance	0.000 sec
69	13:58:51	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'WITHDRAW', ...	1 row(s) affected	0.000 sec

22. PROCEDURE – ACCOUNT TRANSFER

```

316     END IF;
317 END //
318
319 DELIMITER ;
320 • -- Step 1: Call procedure to transfer money
321 CALL TransferMoney(10001, 10002, 5000);
322
323 -- Step 2: Check updated balances for both accounts
324 • SELECT *
325 FROM Account
326 WHERE Account_No IN (10001, 10002);

```

Result Grid

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	51900.00	Hyderabad
10002	102	Savings	75000.00	Vijayawada

Account 8 x

Output

#	Time	Action	Message	Duration / Fetch
66	13:57:34	CREATE TRIGGER PreventInsufficientWithdrawal BEFORE INSERT ON Bank_Transaction FOR EACH ROW...	0 row(s) affected	0.016 sec
67	13:57:42	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'WITHDRAW', ...	Error Code: 1644. Withdrawal failed: Insufficient balance	0.000 sec
68	13:57:49	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'WITHDRAW', ...	Error Code: 1644. Withdrawal failed: Insufficient balance	0.000 sec
69	13:58:51	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'WITHDRAW', ...	1 row(s) affected	0.000 sec
70	14:00:12	CREATE PROCEDURE TransferMoney(IN SenderAccount INT, IN ReceiverAccount INT, IN Transfer...	0 row(s) affected	0.015 sec
71	14:00:23	SELECT * FROM Account WHERE Account_No IN (10001, 10002) LIMIT 0, 1000	2 row(s) returned	0.000 sec / 0.000 sec

23. PROCEDURE – CUSTOMER LOAN DETAILS

```

338     L.Loan_Type,
339     L.Loan_Amount,
340     L.Interest_Rate
341 FROM Customer C
342 JOIN Loan L
343 ON C.Customer_ID = L.Customer_ID
344 WHERE C.Customer_ID = p_Customer_ID;
345 END //
346
347 DELIMITER ;
348 • CALL GetCustomerLoans(101);

```

Customer_Name	Loan_ID	Loan_Type	Loan_Amount	Interest_Rate
Ravi Kumar	501	Home Loan	5000000.00	7.90

Result 9 x

Output

#	Time	Action	Message	Duration / Fetch
68	13:57:49	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'WITHDRAW', ...	Error Code: 1644. Withdrawal failed: Insufficient balance	0.000 sec
69	13:58:51	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'WITHDRAW', ...	1 row(s) affected	0.000 sec
70	14:00:12	CREATE PROCEDURE TransferMoney(IN SenderAccount INT, IN ReceiverAccount INT, IN Transfer...	0 row(s) affected	0.015 sec
71	14:00:23	SELECT * FROM Account WHERE Account_No IN (10001, 10002) LIMIT 0, 1000	2 row(s) returned	0.000 sec / 0.000 sec
72	14:01:09	CREATE PROCEDURE GetCustomerLoans(IN p_Customer_ID INT) BEGIN SELECT C.Customer_N...	0 row(s) affected	0.000 sec
73	14:01:21	CALL GetCustomerLoans(101)	1 row(s) returned	0.000 sec / 0.000 sec

24. PROCEDURE – HIGH BALANCE ACCOUNTS

```

354     IN MinimumBalance DECIMAL(12,2)
355 )
356 BEGIN
357     SELECT *
358 FROM Account
359 WHERE Balance >= MinimumBalance
360 ORDER BY Balance DESC;
361 END //
362
363 DELIMITER ;
364 • CALL HighBalanceAccounts(50000);

```

Account_No	Customer_ID	Account_Type	Balance	Branch
10003	103	Current	120000.00	Bangalore
10005	105	Current	90000.00	Hyderabad
10002	102	Savings	75000.00	Vijayawada
10001	101	Savings	51900.00	Hyderabad

Result 10 x

Output

#	Time	Action	Message	Duration / Fetch
70	14:00:12	CREATE PROCEDURE TransferMoney(IN SenderAccount INT, IN ReceiverAccount INT, IN Transfer...	0 row(s) affected	0.015 sec
71	14:00:23	SELECT * FROM Account WHERE Account_No IN (10001, 10002) LIMIT 0, 1000	2 row(s) returned	0.000 sec / 0.000 sec
72	14:01:09	CREATE PROCEDURE GetCustomerLoans(IN p_Customer_ID INT) BEGIN SELECT C.Customer_N...	0 row(s) affected	0.000 sec
73	14:01:21	CALL GetCustomerLoans(101)	1 row(s) returned	0.000 sec / 0.000 sec
74	14:02:10	CREATE PROCEDURE HighBalanceAccounts(IN MinimumBalance DECIMAL(12,2)) BEGIN SELECT *	0 row(s) affected	0.000 sec
75	14:02:19	CALL HighBalanceAccounts(50000)	4 row(s) returned	0.000 sec / 0.000 sec

25. PROCEDURE WITH IN, OUT PARAMETERS

```

375 INTO p_Balance
376 FROM Account
377 WHERE Account_No = p_Account_No;
378 END //
379
380 DELIMITER ;
381 • -- Step 1: Call procedure passing account number and session variable
382 CALL GetBalance(10001, @CurrentBalance);
383
384 -- Step 2: Print the balance retrieved into @CurrentBalance
385 • SELECT @CurrentBalance;

```

Result Grid | Filter Rows: | Exports: | Wrap Cell Contents: |

@CurrentBalance

Result 11 x | Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
72	14:01:09	CREATE PROCEDURE GetCustomerLoans(IN p_Customer_ID INT) BEGIN SELECT C.Customer_N...	0 row(s) affected	0.000 sec
73	14:01:21	CALL GetCustomerLoans(101)	1 row(s) returned	0.000 sec / 0.000 sec
74	14:02:10	CREATE PROCEDURE HighBalanceAccounts(IN MinimumBalance DECIMAL(12,2)) BEGIN SELECT *	0 row(s) affected	0.000 sec
75	14:02:19	CALL HighBalanceAccounts(50000)	4 row(s) returned	0.000 sec / 0.000 sec
76	14:03:42	CREATE PROCEDURE GetBalance(IN p_Account_No INT, OUT p_Balance DECIMAL(12,2)) BEGIN ...	0 row(s) affected	0.016 sec
77	14:03:50	SELECT @CurrentBalance LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec

26. TRIGGER MANAGEMENT COMMANDS

```

376 FROM Account
377 WHERE Account_No = p_Account_No;
378 END //
379
380 DELIMITER ;
381 • -- Step 1: Call procedure passing account number and session variable
382 CALL GetBalance(10001, @CurrentBalance);
383
384 -- Step 2: Print the balance retrieved into @CurrentBalance
385 • SELECT @CurrentBalance;
386 • SHOW TRIGGERS;

```

Result Grid | Filter Rows: | Exports: | Wrap Cell Contents: |

Trigger	Event	Table	Statement	Timing	Created	sql_mode	Definer	character_set_client	collation_connection
CheckTransactionAmount	INSERT	bank_transaction	BEGIN IF NEW.Amount <= 0 THEN SIGN...	BEFORE	2026-08-14 13:52:29.40	ONLY_FULL_GROUP_BY,STRICT_TRANS_TABLE...	root@localhost	utf8mb4	utf8
PreventInsufficientWithdrawal	INSERT	bank_transaction	BEGIN DECLARE CurrentBalance DECIMAL(12...	BEFORE	2026-08-14 13:57:34.71	ONLY_FULL_GROUP_BY,STRICT_TRANS_TABLE...	root@localhost	utf8mb4	utf8
TransactionAudit	INSERT	bank_transaction	BEGIN INSERT INTO Transaction_Audit (...	AFTER	2026-08-14 13:54:36.27	ONLY_FULL_GROUP_BY,STRICT_TRANS_TABLE...	root@localhost	utf8mb4	utf8
UpdateBalanceAfterTransaction	INSERT	bank_transaction	BEGIN IF NEW.Transaction_Type = 'DEPOSIT...	AFTER	2026-08-14 13:56:31.76	ONLY_FULL_GROUP_BY,STRICT_TRANS_TABLE...	root@localhost	utf8mb4	utf8

Result 12 x | Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
74	14:02:10	CREATE PROCEDURE HighBalanceAccounts(IN MinimumBalance DECIMAL(12,2)) BEGIN SELECT *	0 row(s) affected	0.000 sec
75	14:02:19	CALL HighBalanceAccounts(50000)	4 row(s) returned	0.000 sec / 0.000 sec
76	14:03:42	CREATE PROCEDURE GetBalance(IN p_Account_No INT, OUT p_Balance DECIMAL(12,2)) BEGIN ...	0 row(s) affected	0.016 sec
77	14:03:50	SELECT @CurrentBalance LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
78	14:04:23	DROP TRIGGER CheckBalance	0 row(s) affected	0.015 sec
79	14:04:33	SHOW TRIGGERS	4 row(s) returned	0.000 sec / 0.000 sec

```

380 DELIMITER ;
381 • -- Step 1: Call procedure passing account number and session variable
382 CALL GetBalance(10001, @CurrentBalance);
383
384 -- Step 2: Print the balance retrieved into @CurrentBalance
385 • SELECT @CurrentBalance;
386 • SHOW TRIGGERS;
387 DELIMITER //
388
389 • CREATE TRIGGER CheckBalance
390 BEFORE UPDATE ON Account
391 FOR EACH ROW
392 BEGIN
393     IF NEW.Balance < 0 THEN
394         SIGNAL SQLSTATE '45000'
395         SET MESSAGE_TEXT = 'Transaction failed: Insufficient balance';
396     END IF;
397 END //
398
399 DELIMITER ;

```

Output

#	Time	Action	Message	Duration / Fetch
78	14:04:23	DROP TRIGGER CheckBalance	0 row(s) affected	0.015 sec
79	14:04:33	SHOW TRIGGERS	4 row(s) returned	0.000 sec / 0.000 sec
80	14:04:48	SHOW CREATE TRIGGER CheckBalance	Error Code: 1360. Trigger does not exist	0.000 sec
81	14:04:54	DROP TRIGGER CheckBalance	Error Code: 1360. Trigger does not exist	0.000 sec
82	14:06:48	DROP TRIGGER CheckBalance	Error Code: 1360. Trigger does not exist	0.015 sec
83	14:07:28	CREATE TRIGGER CheckBalance BEFORE UPDATE ON Account FOR EACH ROW BEGIN IF NEW.Bala...	0 row(s) affected	0.000 sec

27. PROCEDURE MANAGEMENT COMMANDS

```

387 DELIMITER //
388
389 • CREATE TRIGGER CheckBalance
390 BEFORE UPDATE ON Account
391 FOR EACH ROW
392 BEGIN
393     IF NEW.Balance < 0 THEN
394         SIGNAL SQLSTATE '45000'
395         SET MESSAGE_TEXT = 'Transaction failed: Insufficient balance';
396     END IF;
397 END //
398
399 DELIMITER ;
400 • SHOW PROCEDURE STATUS
401 WHERE Db = 'BankDB';
402
403
404 • DROP PROCEDURE GetAllCustomers;
405
406

```

Output

#	Time	Action	Message	Duration / Fetch
79	14:04:33	SHOW TRIGGERS	4 row(s) returned	0.000 sec / 0.000 sec
80	14:04:48	SHOW CREATE TRIGGER CheckBalance	Error Code: 1360. Trigger does not exist	0.000 sec
81	14:04:54	DROP TRIGGER CheckBalance	Error Code: 1360. Trigger does not exist	0.000 sec
82	14:06:48	DROP TRIGGER CheckBalance	Error Code: 1360. Trigger does not exist	0.015 sec
83	14:07:28	CREATE TRIGGER CheckBalance BEFORE UPDATE ON Account FOR EACH ROW BEGIN IF NEW.Bala...	0 row(s) affected	0.000 sec
84	14:07:59	DROP PROCEDURE GetAllCustomers	0 row(s) affected	0.000 sec